CHAPTER 3

Mandibular Denture Suction in conjunction with BPS

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1 You are the witness

"It is difficult to achieve a suction effect for the mandibular complete denture using the border-molding technique with impression compound taught in dental schools." [8]

The readers will prove this to themselves when they try to border mold using compound.

Why do we fail to create a suction effect with the conventional technique? What are the factors causing a loss of suction in the denture fabrication process?

Without knowing the answers to these questions, the mandibular complete dentures we fabricate will always have the problem of lifting, no matter how much effort we put in.

This issue can be solved by integrating the mechanism of mandibular denture suction into the BPS denture fabricating system and incorporating the integrated approach into the practice of complete denture treatment.

2 Suction leads to success in mandibular complete denture treatment

Generally, the success of the prosthetic treatment of edentulous jaws depends on the clinician's skills and the status of the patient's stomatognathic system. Despite the continued deterioration of residual ridge conditions seen among edentulous patients in recent years, an increasing number of reports show that the level of patient satisfaction improves with suction-effective mandibular complete dentures [6,7]. Has the suction effect been achieved by chance? Is there a special technique or secret for creating suction?

In either case, the following is true. Once the clinician succeeds in creating suction for the mandibular complete denture, he/she will no longer want to return to the conventional denture fabrication technique using impression compound, because both the patient and clinician "can feel the difference." [8–10]

3 Differences in the concept of retention between conventional and suction-effective dentures

Let's examine two different final impressions taken for the same patient. The upper diagram shows one made with the traditional impression compound technique introduced by Zarb et al. in 1970 [8], see "Prosthodontic Treatment for Edentulous Patients," and is taught in dental schools to date (**Fig 3-1**). The lower diagram depicts the functional impression using the custom tray derived from the preliminary impression technique devised by the author based on the suction mechanism, which will be described later (**Fig 3-2**).

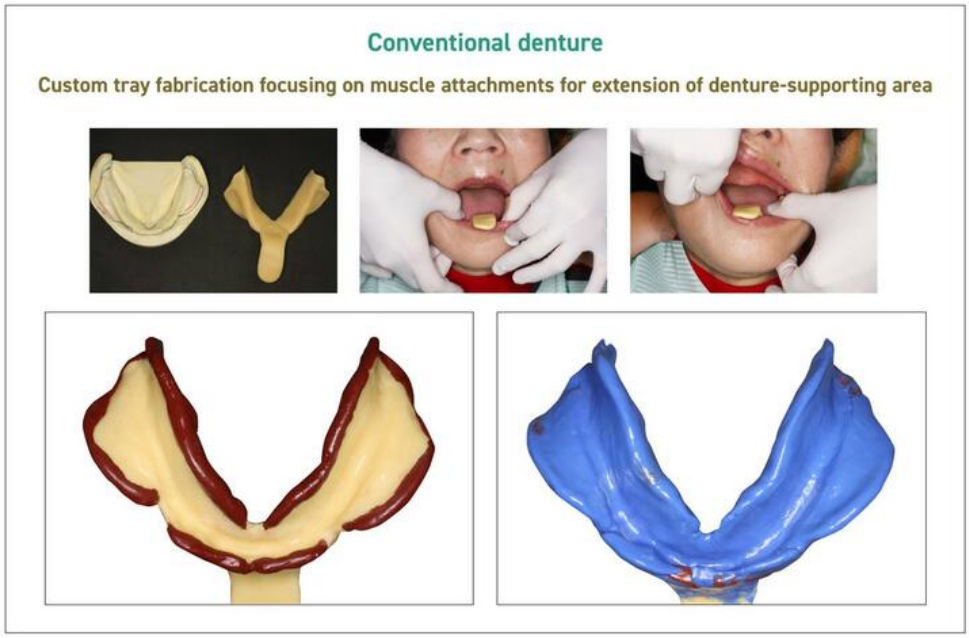


Fig 3-1 The conventional technique was taught in dental school. The aim of conventional denture is to maximize a denture-supporting area and bring a wide interior surface of denture base into intimate contact with alveolar ridge, rather than to create a suction effect.

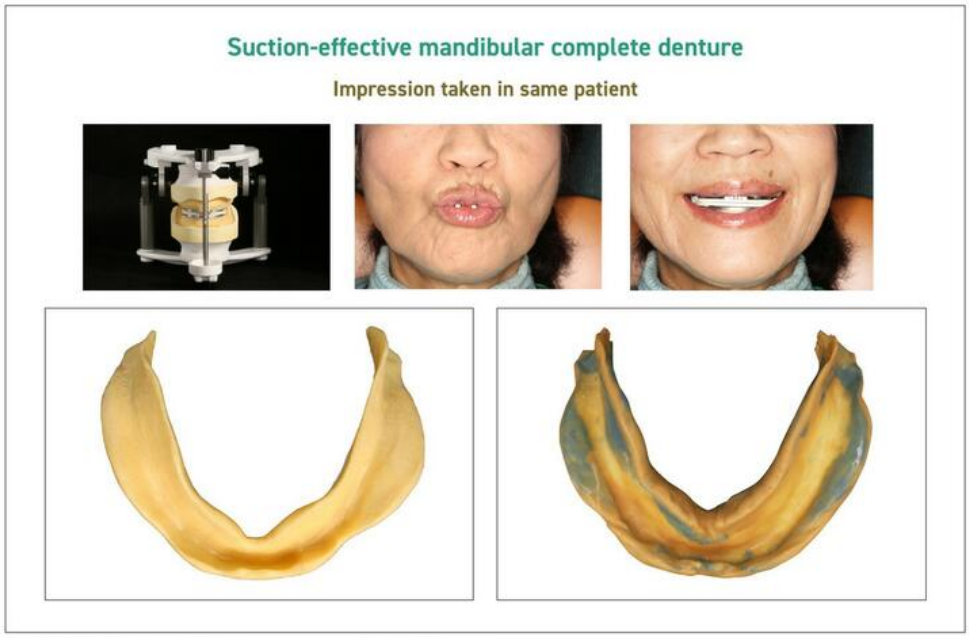


Fig 3-2 Entire borders of denture base surrounded by oral mucosa.

The conventional denture fabrication aims to increase the stress-bearing area of the denture base, bring a wide area of its interior surface into intimate contact with the alveolar ridge, and thereby obtain good denture retention and stability for improved chewing function. Conventional dentures of this type are therefore called retention and stabilization-type dentures.

In contrast, the "suction-effective denture" shown in the lower diagram of **Fig 3-2** is designed to create negative pressure temporarily during swallowing or chewing by creating an effective seal around the entire border of the denture base and the mobile mucosa. This seal prevents the denture from lifting during speaking and reinforces denture retention and stability during function. The operator feels resistance due to negative pressure when removing the denture. This sensation is called "suction," a state of enhanced retention by negative pressure added to the conventional retention and stabilization mechanism (**Fig 3-3**).

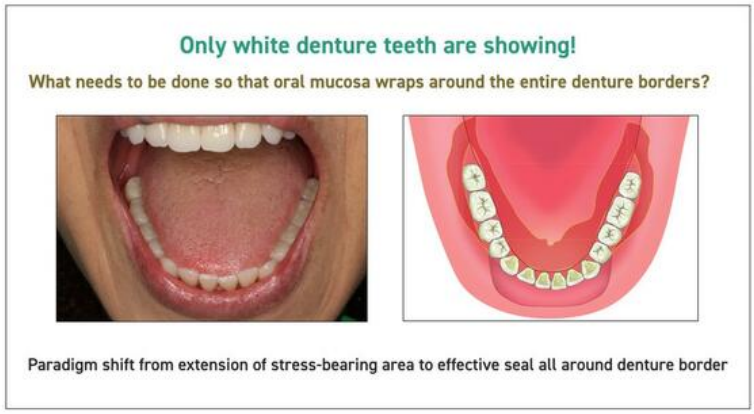


Fig 3-3 Mandibular denture suction can be accomplished with the sealing of the entire borders by mobile mucosa, resulting in display of only white denture teeth.

This difference in the concept of retention between the conventional and suction-effective dentures translates into significant differences in the fabrication process (**Fig 3-4**). To achieve suction for the mandibular complete denture, we must first understand the mechanism of suction. The next requirement is to use appropriate materials and systems for denture fabrication. BPS is currently the best denture fabrication system that offers the highest quality, ease of leaning and best chance of success. What appears to be a subtle difference in denture fabrication concept will come out as a massive difference in treatment outcome. To achieve suction or not to achieve suction – that is a serious question for us clinicians (**Fig 3-4**).

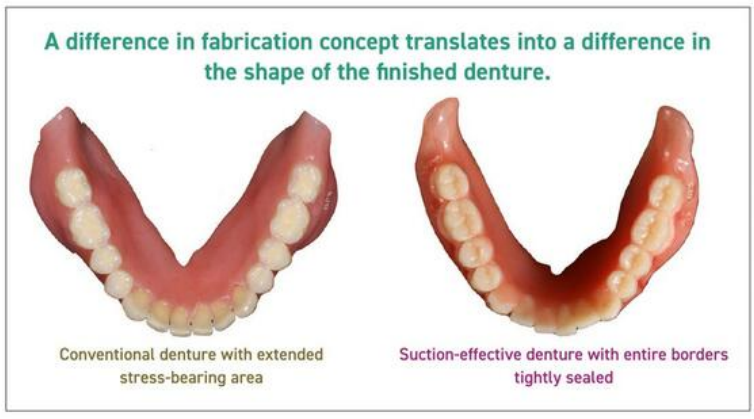


Fig 3-4 (Finished dentures for same patient) A subtle difference in denture fabrication concept will come out as major difference in treatment outcome. For clinicians, success or failure in obtaining a good suction with the mandibular denture makes a world of difference.

4 Do-able impression technique that everyone can achieve

A technique to fabricate a mandibular suction-effective denture must be simple so that anyone interested can readily acquire and implement this method, and that everyone can capture the same anatomical information critical for suction in the impression. How can this objective be accomplished?

The conventional compound impression technique involves border molding by the operator, which is affected by his/her experience level. The same anatomical information must be captured for the same mouth; whoever takes the impression. However, the compound impression is influenced by the operator's clinical experience and skill level, indicating that it is not a simple, unfailing impression technique that everyone can achieve.

The impression technique that meets these criteria is a functional impression technique based on functional movements performed by the patient. The functional impression technique of BPS satisfies this extremely important requirement.

5 Raise your skill level with "Suction Mechanism"

Factors leading to success in complete denture treatment are categorized into two groups;

1) one related to the conditions of the patient's stomatognathic system, such as the status of alveolar ridge resorption and neuromuscular factors involved in occlusion; and
2) the other related to clinical denture fabrication procedures including impression-taking, bite registration, the arrangement of prosthetic teeth and resin polymerization.

The two most important factors are occlusion and impression taking. No matter how good the operator is at taking impressions, denture stability would be compromised if there is a discrepancy between the mandibular jaw position established by the operator and the patient's true occlusal position. Conversely, even with a proper occlusion, the denture will lift upon mouth opening if the operator is lacking in impression-taking skills.

BPS has integrated these two important pillars into one. There is no doubt that dentists who understand and implement the basics of BPS will be able to successfully incorporate the suction mechanism for the mandibular complete denture into BPS to enhance their clinical standards.

6 Why is suction easier to obtain for the maxillary denture and more difficult for the mandibular denture?

Reasons, why it is more difficult to achieve retention, stability and suction of the mandibular denture, are

1) Presence of the tongue
2) A smaller area of the interior surface of the denture base in contact with the underlying mucosa.
3) A smaller amount of immobile ridge mucosa (masticatory mucosa)
4) Deformable retromolar pads
5) Two to three times larger movable volume of mucobuccal and mucolingual fold [11]

The 4th reason is little-known, but it would not be difficult to understand if we stop and think about how the posterior buccal area is border-molded. Let's briefly review the border molding of the posterior buccal area with compound.

For the maxilla, when the operator softens the impression compound and retracts the cheeks and lips with his/her fingers, little change takes place in the shape of the buccal border beyond a certain point. In the mandible, however, the buccal border continues to move inward and decreases in size with repeated border molding. The difficulty encountered in the mandible is caused by a larger movable volume of the mucobuccal fold than in the maxilla (**Fig 3-5**). When should the operator stop bolder molding in the mandible? With how much force and in what direction should the cheeks be retracted to obtain an optimal shape of the border? Pursuit of perfection in border molding would force the operator into the realm of craftsmanship that requires experience and skills.

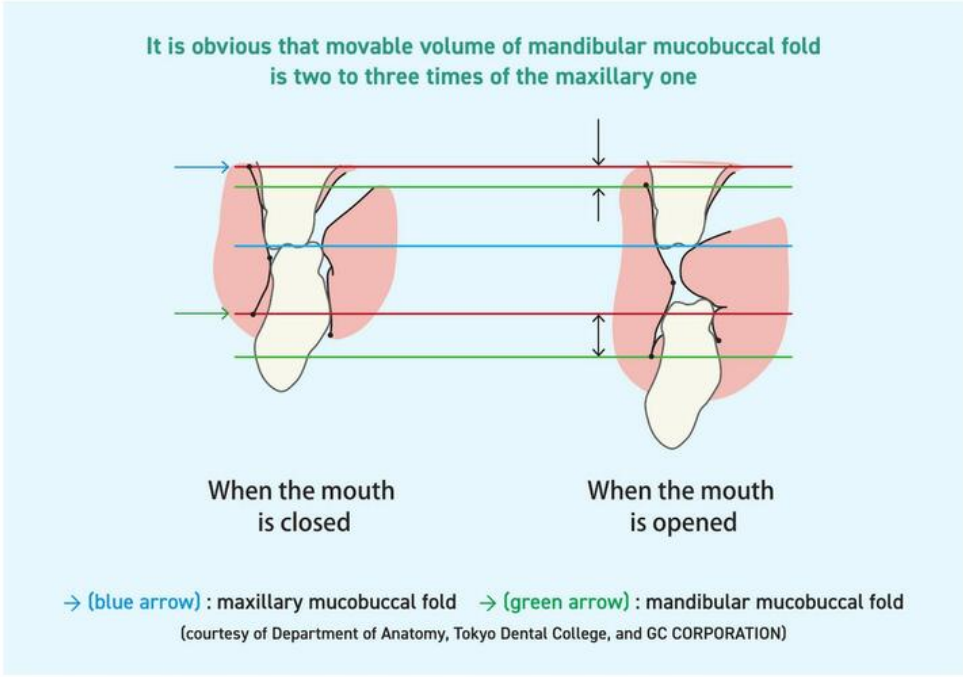


Fig 3-5 Difference in the movable volume of mucobuccal fold between the maxilla and mandible.